Performance & Testing Report

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> **Implemented fixes verified:** 229 checks passed in Chromium 144.0.7559.96. The table, menu, form, themes, project rendering, and image decoding passed the tested local scenarios.

Test environment

Playwright controlled a headless Chromium browser at 375 x 900, 768 x 900, and 1280 x 900 pixels in both themes. The test document was about:blank, populated from the submitted source. Localhost navigation was blocked in this environment, so CSS, JavaScript, and JPG bytes were embedded only in the temporary test fixture.

The delivered website still uses index.html, external style.css and script.js, and relative image paths. This method checks local behavior; it does not measure a published page or network loading.

Performance issue and optimization

The uploaded profile PNG was substantially larger than needed for the portfolio. The website uses an optimized JPG at the same 575 x 577 dimensions. File measurements confirm the reduction:

Image version	Size	Comparison
Uploaded profile PNG	387,768 bytes	Source image
Earlier portfolio JPG	41,206 bytes	Earlier package
Revised profile.jpg	35,642 bytes	13.5% below earlier JPG

The revised profile JPG is **90.8% smaller** than the uploaded PNG. The site image retains explicit dimensions and high fetch priority. These values are file-size measurements, not load-time or Lighthouse performance scores.

Additional implementation checks

All six submitted images are genuine JPEGs with the .jpg extension. The three project images use lazy loading and explicit dimensions. The CSS includes fluid image sizing, mobile-first breakpoints, and reduced-motion rules. No external fonts or third-party runtime libraries are loaded.

The HTML, CSS, JavaScript, and four website JPGs total **151,657 bytes on disk**, excluding wireframes and reports. This is an asset-size total, not a network transfer measurement.

Verification Results

Completed checks and the scope of the evidence

Test group	Checks passed	Table / container
375 px - dark and light	80 / 80	337 / 337 px
768 px - dark and light	80 / 80	730 / 730 px
1280 px - dark and light	62 / 62	1,118 / 1,118 px
Source and image checks	7 / 7	Formats and implementation
Total	229 / 229	No failed checks

Behavior confirmed

Tests exercised the five sections, three generated project cards, theme switching, responsive navigation, Escape-to-close and focus return, touch and keyboard actions, resize state, the visible skip link, form validation, and image decoding. No unintended page-wide horizontal overflow or uncaught JavaScript errors appeared in the tested states.

Requirement reference

COP 3835 Final Project instructions: Parts 3E, 7G, 8A-8G, and 9F